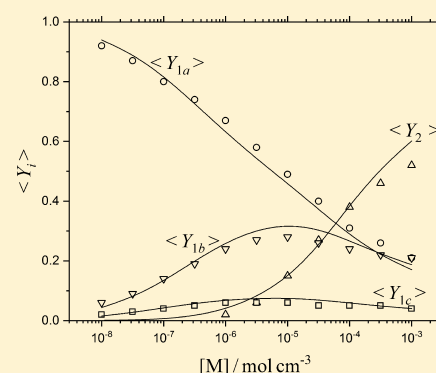


Simplified Analysis and Representation of Multichannel Thermal Unimolecular Reactions

Jürgen Troe*

Institut für Physikalische Chemie, Universität Göttingen, Tammannstrasse 6, D-37077 Göttingen, Germany
Max-Planck-Institut für Biophysikalische Chemie, Am Fassberg 11, D-37077 Göttingen, Germany

ABSTRACT: Two-channel and multichannel thermal unimolecular reactions are analyzed by simple models, starting with the calculation of separated-channel rate constants and accounting for intrinsic channel coupling afterward. Reactions with rigid- and with loose-activated complex channels are distinguished. Weak-collision, energy-transfer, effects are suggested to govern the competition between rigid-activated complex channels, while angular-momentum, “rotational channel switching”, effects dominate the competition between rigid- and loose-activated complex channels. The models are tested against master equation treatments of the dissociations of formaldehyde and of glyoxal from the literature. Besides giving insight into the influence of various molecular input parameters, the present approach leads to compact representations of rate constants suitable for inclusion in databases.



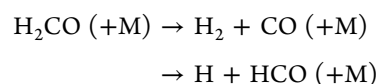
1. INTRODUCTION

Multichannel reactions are ubiquitous in chemical kinetics. Often, these processes involve complicated intrinsic dynamics and are difficult to describe in a transparent manner. As a consequence, a compact representation of their temperature and pressure dependence is not easily obtained. Such representations, however, appear desirable when the reactions are to be included in databases (such as, e.g., refs 1–3 for combustion and atmospheric chemistry). The situation obviously is more complicated than that for single-channel reactions where compact, approximate, representations of the rate constants in terms of a few characteristic rate parameters are in use. Rate constants of single-channel reactions can, for instance, be formulated in extended Lindemann–Hinshelwood form, employing limiting low-pressure and high-pressure rate constants together with “broadening factors” accounting for the finer details of the falloff interpolation between the limiting low- and high-pressure ranges (see refs 4–7 and alternative references cited therein).

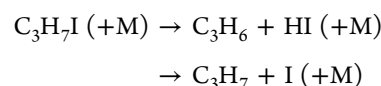
The present work tries to extend the simplified representation of single-channel thermal unimolecular reactions to multichannel systems. One may start, e.g., with a description of decoupled reaction channels treated by single-channel unimolecular rate theory. The difficulty then, however, consists of accounting for the intrinsic coupling of the channels. The situation is similar to complex-forming bimolecular reactions, for which channel coupling has been expressed in terms of falloff representations of single-channel unimolecular reactions and their reverse processes (see refs 8 and 9 for the reaction $C_2H_5 + O_2$ and ref 10 for the reaction $CH_3OCH_3 + Fe^+$). In an analogous way, starting from decoupled single-channel representations and accounting for channel coupling afterward, the present work deals with two-channel thermal unimolecular

reactions and, finally, with a generalization to multichannel systems.

Two features of the dynamics have to be considered in particular. Under low-pressure conditions, the energetically lowest channel (threshold energy E_{01}) usually is more strongly favored over higher channels than in the corresponding single-channel reactions. Collisional energy transfer under these conditions does not populate states with energies larger than E_{01} to the extent achieved in single-channel reactions under similar conditions. Furthermore, an effect named “rotational channel switching” may occur^{11,12} when the individual channels have different angular momentum (quantum number J) dependences of their threshold energies $E_{0i}(J)$. This may happen when the energetically more favored channels (at $J = 0$) are of elimination character and have rigid activated complexes (ACs), while the energetically less favored channels (at $J = 0$) are of bond fission type and have loose ACs. Examples for this behavior are the two-channel dissociations of formaldehyde (see, e.g., refs 13–15)



and of alkyl iodides such as¹⁶

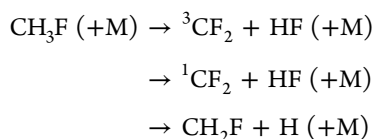


or the multichannel dissociations of methyl fluoride^{17,18}

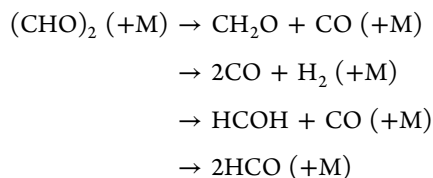
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and of glyoxal¹⁹



Detailed master equation (ME) simulations accounting for weak collision and angular momentum effects are available for these systems. The corresponding results may serve as tests for the simplified analysis proposed in the present Article.

ME calculations for two-channel reactions in ref 20 showed that the branching fraction of the upper channel (threshold energy E_{02}) under low-pressure conditions contains the factor $\exp[-(E_{02} - E_{01})/\gamma]$, where γ is the average energy transferred per up-collision in an exponential collision model.²¹ This factor disappears in the high-pressure limit where decoupled- and coupled-channel rate constants approach each other. In addition to the collisional factor, details of the specific rate constants $k_i(E)$ for unimolecular reaction of the two channels enter the branching fraction. In particular, the J dependence of the threshold energies $E_{0i}(J)$ plays an important role, as it may change the energetic order of the threshold energies when rotational channel switching occurs.

The following Article starts with a simplified analysis of rate constants for decoupled- and coupled-channel reactions in the framework of a simple Lindemann–Hinshelwood type, strong collision, model.²⁰ The results then are improved by refining the limiting rate constants which are used as input quantities. In this respect, the concept follows that employed for single-channel reactions.⁴ After this, it elaborates how results from decoupled-, separated-channel, calculations can be extended to account for channel coupling. Weak collision and angular momentum effects then are included in the limiting low- and high-pressure rate constants, while falloff interpolations between these limits are first represented in simplified Lindemann–Hinshelwood form and more realistic broadening factors are introduced afterward. Rotational channel switching and specificities of the potential energy surfaces (PESs) are introduced in a transparent way. The ME calculations from refs 13 and 19 provided Chebychev polynomial representations of the rate constants. The present approach sacrifices the accuracy obtained in these references against simplicity. In this way, it becomes particularly suitable for inclusion in databases. Furthermore, the consequences of uncertainties in the molecular input parameters can be traced easily. A generalization from two-channel to multichannel reactions is proposed at the end.

2. REDUCED REPRESENTATION OF THE PRESSURE DEPENDENCE OF TWO-CHANNEL REACTIONS

Single-channel thermal unimolecular reactions in ref 4 were represented in doubly reduced form by expressing the ratio of the pseudo-first-order rate constants k to the limiting high-pressure rate constants k_∞ as a function of “reduced pressures” (or “reduced bath gas concentrations”), given by the ratio of

the extrapolated, limiting pseudo-first-order, low-pressure rate constants k_0 (being proportional to the bath gas concentration $[\text{M}]$ or to the pressure P) to k_∞ . The “center of the falloff curves” then was located at those values of $[\text{M}]$ or P (denoted by $[\text{M}]_{\text{cent}}$ or P_{cent}), for which $k_0/k_\infty = 1$. In this doubly reduced representation, k/k_∞ takes the form

$$k/k_\infty = \text{LH}(x) F(x) \quad (1)$$

where $x = k_0/k_\infty = [\text{M}]/[\text{M}]_{\text{cent}} = P/P_{\text{cent}}$, the Lindemann–Hinshelwood factor $\text{LH}(x) = x/(x+1)$, and some broadening factor $F(x)$ ^{4–7} whose most important feature is its center value $F_{\text{cent}} = F(x=1)$. For simplicity, we use the original approximation⁴ of $\log F(x) \approx [1 + ((\log x)/N)^2]^{-1} \log F_{\text{cent}}$ with $N \approx 0.75 - 1.27 \log F_{\text{cent}}$ instead of the more advanced form of $F(x)$ from refs 6 and 7. In the following, in an analogous way, we determine falloff curves for two-channel reaction systems, first by neglecting and later by introducing channel coupling effects.

A Lindemann–Hinshelwood-type, strong-collision, approach to two-channel reactions has been formulated before.²⁰ Neglecting J dependences, stepwise energy independent specific rate constants $k_i(E)$ for the two channels (subscripts 1 and 2) were employed (with channel 1 being energetically more favorable than channel 2). $k_1(E) = K_1$ and $k_2(E) = 0$ were used for $E_{01} \leq E \leq E_{02}$, while $k_1(E) = K_1^*$ and $k_2(E) = K_2$ were taken for $E \geq E_{02}$. The coupled-channel rate constants then follow as

$$k_1 = \frac{Z[\text{M}]K_1}{K_1 + Z[\text{M}]} \int_{E_{01}}^{E_{02}} f(E) dE + \frac{Z[\text{M}]K_1^*}{K_1^* + K_2 + Z[\text{M}]} \int_{E_{02}}^{\infty} f(E) dE \quad (2)$$

$$k_2 = \frac{Z[\text{M}]K_2}{K_1^* + K_2 + Z[\text{M}]} \int_{E_{02}}^{\infty} f(E) dE \quad (3)$$

(Z denotes the collision number, $[\text{M}]$ the bath gas concentration, and $f(E)$ the equilibrium population of the vibrationally excited states of the reactant; in the following, we use the abbreviations $\int_{E_{01}}^{E_{02}} f(E) dE = F_1$ and $\int_{E_{02}}^{\infty} f(E) dE = F_2$).

Equations 2 and 3 may be exploited in a simple way. At first, a channel coupling is neglected and “separated-channel rate constants” $k_i^{\text{s.ch.}}$ (superscripts s.ch.) are derived by putting $K_2 = 0$ for $k_1^{\text{s.ch.}}$ and $K_1 = K_1^* = 0$ for $k_2^{\text{s.ch.}}$. This leads to separated-channel falloff curves of the $k_i^{\text{s.ch.}}$ whose centers are located at $[\text{M}]_{i,\text{cent}}^{\text{s.ch.}} = k_{i,\infty}^{\text{s.ch.}}/(k_{i,0}^{\text{s.ch.}}/[\text{M}])$. After this, the centers of the coupled-channel (no superscripts) falloff curves are derived from $[\text{M}]_{i,\text{cent}} = k_{i,\infty}/(k_{i,0}/[\text{M}])$. This leads to

$$Z[\text{M}]_{2,\text{cent}}^{\text{s.ch.}} = K_2 \quad (4)$$

and

$$Z[\text{M}]_{2,\text{cent}} = K_1^* + K_2 \quad (5)$$

Likewise, one has

$$Z[\text{M}]_{1,\text{cent}}^{\text{s.ch.}} = (K_1 F_1 + K_1^* F_2)/(F_1 + F_2) \quad (6)$$

and

$$Z[\text{M}]_{1,\text{cent}} = (K_1 F_1 + K_1^* F_2)/(F_1 + K_1^* F_2/[K_1 + K_1^*]) \quad (7)$$

The shifts from $[\text{M}]_{i,\text{cent}}^{\text{s.ch.}}$ to the larger values $[\text{M}]_{i,\text{cent}}$ are the result of channel coupling. As the $k_{i,\infty}^{\text{s.ch.}}$ and $k_{i,\infty}$ are identical,

the shifts are caused by the difference in the $k_{i,0}^{\text{s.ch.}}$ and $k_{i,0}$. The center of the falloff curve for the total rate constant $k_{\text{tot}} = k_1 + k_2$ is shifted from $Z[M]_{\text{tot,cent}}^{\text{s.ch.}} = (K_1 F_1 + [K_1^* + K_2] F_2) / (F_1 + 2F_2)$ to

$$Z[M]_{\text{tot,cent}} = (K_1 F_1 + ([K_1^* + K_2] F_2) / (F_1 + F_2)) \quad (8)$$

Analogous to the $[M]_{i,\text{cent}}$ $[M]_{\text{tot,cent}}$ is also larger than $[M]_{\text{tot,cent}}^{\text{s.ch.}}$. It appears noteworthy that the falloff curves for k_1 , k_2 , and k_{tot} in this simple approach are all of the form of eq 1 with broadening factors $F(x)$ equal to unity.

We next consider the branching fraction Y_2 for the upper channel, defined by

$$Y_2 = k_2 / (k_1 + k_2) \quad (9)$$

The described approach now leads to

$$Y_2 = Y_{2,0,0} + (Y_{2,\infty} - Y_{2,0,0})x / (x + 1) \quad (10)$$

where $x = [M] / [M]_{Y_{2,\text{cent}}}$ and

$$Y_{2,\infty} = K_2 F_2 / (K_1 F_1 + [K_1^* + K_2] F_2) \quad (11)$$

The pressure independent branching fraction at $[M] \rightarrow 0$ follows as

$$Y_{2,0,0} = K_2 F_2 / ([K_1^* + K_2] F_1 + F_2) \quad (12)$$

while $Z[M]_{Y_{2,\text{cent}}}$ takes a more complicated form which is not reproduced here in detail. Expressing Y_2 as a function of the ratio $V_M = Z[M] / (K_1^* + K_2)$, expanding the result into a power series, and retaining only terms of first order in V_M , however, one obtains

$$Z[M]_{Y_{2,\text{cent}}} \approx K_1 (F_1 + F_2) (K_1^* + K_2) / (K_1 F_1 + [K_1^* + K_2] F_2) \quad (13)$$

Apart from the nonzero value at $[M] \rightarrow 0$, Y_2 is again of the form of eq 1 with a broadening factor equal to unity. For the separated-channel rate constants, one has $Y_{2,0,0}^{\text{s.ch.}} = F_2 / (F_1 + F_2)$ and $Y_{2,\infty}^{\text{s.ch.}} = Y_{2,\infty}$. $Z[M]_{Y_{2,\text{cent}}}^{\text{s.ch.}}$ in an analogous way can also be expressed by the parameters K_1 , K_1^* , K_2 , F_1 , and F_2 , but it again takes a more complicated form which is not reproduced here in an analogous way.

The parameters K_1 , K_1^* , K_2 , F_1 , and F_2 can be related to the separated-channel rate constants $k_{1,0}^{\text{s.ch.}}$, $k_{1,\infty}^{\text{s.ch.}}$, $k_{2,0}^{\text{s.ch.}}$, and $k_{2,\infty}^{\text{s.ch.}}$. However, in order to be complete, one additional parameter is required which reflects the extent of channel coupling. $k_{2,0}$ can be used for this purpose. The relationships required for the determination of the parameters K_1 , K_1^* , K_2 , F_1 , and F_2 then are $k_{1,0}^{\text{s.ch.}} = Z[M] (F_1 + F_2)$, $k_{1,\infty}^{\text{s.ch.}} = K_1 F_1 + K_1^* F_2$, $k_{2,0}^{\text{s.ch.}} = Z[M] F_2$, $k_{2,\infty}^{\text{s.ch.}} = Z[M] K_2 F_2 / (K_1^* + K_2)$, and $k_{2,\infty}^{\text{s.ch.}} = K_2 F_2$. These parameters also lead to $Y_{2,0,0}$, $Y_{2,\infty}$, and $Z[M]_{Y_{2,\text{cent}}}$ which are given by

$$Y_{2,0,0} = k_{2,0} / k_{1,0}^{\text{s.ch.}} \quad (14)$$

$$Y_{2,\infty} = k_{2,\infty}^{\text{s.ch.}} / (k_{1,\infty}^{\text{s.ch.}} + k_{2,\infty}^{\text{s.ch.}}) \quad (15)$$

$$Z[M]_{Y_{2,\text{cent}}} \approx (Y_{2,\infty} / Y_{2,0,0}) (k_{1,\infty}^{\text{s.ch.}} - k_{2,\infty}^{\text{s.ch.}} [1 - G] / G) / ([k_{1,0}^{\text{s.ch.}} - k_{2,0}^{\text{s.ch.}}] / Z[M]) \quad (16)$$

and $k_{\text{tot},0}$, $k_{\text{tot},\infty}$, and $Z[M]_{\text{tot,cent}}$ which are given by

$$k_{\text{tot},0} = k_{1,0}^{\text{s.ch.}} \quad (17)$$

$$k_{\text{tot},\infty} = (k_{1,\infty}^{\text{s.ch.}} + k_{2,\infty}^{\text{s.ch.}}) \quad (18)$$

$$Z[M]_{\text{tot,cent}} = (k_{1,\infty}^{\text{s.ch.}} + k_{2,\infty}^{\text{s.ch.}}) / (k_{1,0}^{\text{s.ch.}} / Z[M]) \quad (19)$$

G here denotes $G = K_2 / (K_1^* + K_2) = k_{2,0} / k_{2,\infty}^{\text{s.ch.}}$. If separated-channel calculations for $k_1^{\text{s.ch.}}$ and $k_2^{\text{s.ch.}}$ have been made (and $k_{2,0}$ can be estimated as shown below), eqs 9–19 can finally be used to represent the branching fractions Y_2 and k_{tot} . Likewise, the coupled-channel rate constants k_1 and k_2 from eqs 2 and 3 can be expressed in terms of $k_1^{\text{s.ch.}}$, $k_2^{\text{s.ch.}}$, and $k_{2,0}$.

The given approach suggests that the transition of the branching fraction Y_2 from a low-pressure limiting value $Y_{2,0,0}$ to a high-pressure limiting value $Y_{2,\infty}$ can be represented in the form of eq 10 (Figure 3 of ref 20 shows numerical examples which correspond to the model elaborated here in analytical form). However, a number of questions arise: (i) Would eq 16 be sufficient for an estimate of the center $[M]_{Y_{2,\text{cent}}}$ of the transition curve of Y_2 under more realistic conditions? (ii) Would broadening factors $F(x = [M] / [M]_{Y_{2,\text{cent}}})$ different from unity be required in eq 10? (iii) Would the separated-channel rate constants $k_1^{\text{s.ch.}}$ and $k_2^{\text{s.ch.}}$ be sufficient for the estimate of $Y_{2,0,0}$, $Y_{2,\infty}$, and $Z[M]_{Y_{2,\text{cent}}}$ by means of eqs 14–16? The answer to question iii is clearly “no”, because the treatment given so far applies to a single J only, while $k_1^{\text{s.ch.}}$ and $k_2^{\text{s.ch.}}$ generally are obtained as rotationally averaged quantities. If there is rotational channel switching, the order of the two channels changes: the rigid-AC channel will correspond to channel 1 only at low J , while it corresponds to channel 2 at high J . Averaging over J thus needs to be performed, as is generally done for $k_1^{\text{s.ch.}}$ and $k_2^{\text{s.ch.}}$. Questions i and ii can only be answered by comparison with detailed ME calculations; see the following. If no further theoretical analysis is available, $Y_{2,0,0}$, $Y_{2,\infty}$, and $Z[M]_{Y_{2,\text{cent}}}$ at least can be employed as empirical fit parameters for a representation of experimental values of $Y_2([M])$. This is demonstrated in the following sections.

3. WEAK COLLISION AND ANGULAR MOMENTUM EFFECTS

The simple representation of total rate constants k_{tot} and branching fractions Y_2 in terms of separated-channel rate constants described in section 2 has to be extended to account for weak collision and angular momentum effects. One could do this by employing the reduced, three-parameter, representation of section 2 and modifying the input parameters only. These parameters are $k_{\text{tot},0}$, $k_{\text{tot},\infty}$, and $F_{\text{tot,cent}}$ for k_{tot} as well as $Y_{2,0,0}$, $Y_{2,\infty}$, possibly $F_{Y_{2,\text{cent}}}$, and $Z[M]_{Y_{2,\text{cent}}}$ for Y_2 . Obviously, $Y_{2,\infty}$ and $k_{\text{tot},\infty}$ follow directly from the corresponding separated-channel quantities given in eqs 15 and 18. On the other hand, $k_{\text{tot},0}$ could be calculated in the framework of single-channel unimolecular rate theory using E_{01} as the threshold energy. However, rotational channel switching now leads to a modified “rotational factor” F_{rot} in $k_{\text{tot},0}$.⁴ This problem has been considered for the dissociation of formaldehyde in ref 15. Only when it is properly handled, the analysis of the experimental $k_{\text{tot},0}$ allows one to deduce an average energy transferred per collision $\langle \Delta E \rangle$ which is consistent between thermal dissociation¹⁵ and photodissociation experiments.²² Often, both in the analysis of $k_{\text{tot},0}$ and in ME calculations of multichannel rate constants, $\langle \Delta E \rangle$ is used as an empirical fit parameter. Its uncertainty then constitutes one of the major limitations of the analysis.

The parameters representing k_{tot} and Y_2 in eqs 14–19 depend on the type of the considered reaction. Therefore, in the following, two types of reactions have to be distinguished. We first consider two-channel reactions without rotational channel switching. We name these reactions “type-A reactions”. They are characteristic for situations where the two channels both have rigid ACs with no (or only weak) J dependences of their threshold energies (see, e.g., the systems used for “collisional spectroscopy” in ref 23). The situation is quite different for “type-B reactions” where channel 1 has a rigid AC and channel 2 a loose AC. In the latter case, the $E_{0i}(J)$ have strongly differing J dependences such that $E_{01}(J) < E_{02}(J)$ for $J < J_{\text{sw}}$ and $E_{01}(J) > E_{02}(J)$ for $J > J_{\text{sw}}$; i.e., rotational channel switching occurs at $J = J_{\text{sw}}$. After discussing type-A and type-B, two-channel, reactions, we try to extend the results to selected multichannel reactions.

3.1. Weak Collision Effects in Type-A, Two-Channel, Reactions. Weak collision effects require the solution of a ME. We take advantage of the analytical ME solution from ref 20 which was obtained with the exponential collision model of ref 21. The average energy transferred per down-collision here was denoted by α , while γ characterizes up-collisions. The total energy transferred per collision then is given by $\langle \Delta E \rangle = \gamma - \alpha$. The quantities α and γ are linked by detailed balancing

$$\gamma \approx \alpha F_E k_B T / (\alpha + F_E k_B T) \quad (20)$$

(with the factor F_E accounting for the energy dependence of the vibrational density of states $\rho(E)$ of the reactant).²¹ Besides the collision model, also details of the specific rate constants $k_i(E)$ of the two channels need to be known and the Lindemann–Hinshelwood form of the specific rate constants used in section 2 has to be abandoned. With an E -specific branching ratio

$$V_2(E) = k_2(E) / [k_1(E) + k_2(E)] \quad (21)$$

the treatment of ref 20 led to a limiting low-pressure branching fraction

$$\begin{aligned} Y_{2,0,0} &= k_{2,0} / k_{\text{tot},0} \\ &= \exp[-(E_{02} - E_{01})/\gamma] \times \int_{E_{02}}^{\infty} V_2(E) \\ &\quad \exp[-(E_{02} - E_{01})/\gamma] dE/\gamma \end{aligned} \quad (22)$$

Unimolecular rate theory expresses the $k_i(E)$ by $W_i(E)/h\rho(E)$, where the $W_i(E)$ denote the “numbers of AC states”. $V_2(E)$ then is given by

$$V_2(E) = W_2(E) / [W_1(E) + W_2(E)] \quad (23)$$

If channels 1 and 2 have not too different threshold energies and similarly rigid ACs, $W_1(E)$ and $W_2(E)$ over the energy range γ above E_{01} may still be at their threshold values of unity such that $V_2(E)$ is at its threshold value $1/2$. This would lead to

$$Y_{2,0,0} \approx 0.5 \exp[-(E_{02} - E_{01})/\gamma] \quad (24)$$

If stronger energy dependences of $W_1(E)$ and $W_2(E)$ have to be accounted for and $V_2(E)$ is approximated by an exponential expression of the form

$$\begin{aligned} V_2(E) &\approx V_2(E_{02}) + [V_2(\infty) - V_2(E_{02})] \\ &\quad \exp[-(E - E_{02})/\delta] \end{aligned} \quad (25)$$

one would obtain²⁰

$$\begin{aligned} Y_{2,0,0} &\approx [V_2(\infty) + (V_2(E_{02}) - V_2(\infty))\delta/(\delta + \gamma)] \\ &\quad \exp[-(E_{02} - E_{01})/\delta] \end{aligned} \quad (26)$$

One should remember that $k_{\text{tot},0}$ in eq 17 corresponds to calculations without rotational averaging, i.e., to $J = 0$. $k_{2,0}/k_{\text{tot},0}$ thus also only corresponds to $Y_{2,0,0}$ for $J = 0$. The calculation of a rotationally averaged $k_{\text{tot},0}$ within the approach of ref 4, however, requires the true rotational factor F_{rot} . This factor generally is not too different from unity for type-A reactions. As $k_{\text{tot},\infty}$ and $Y_{2,\infty}$ follow directly from the separated-channel rate constants, the remaining parameters to be specified are the broadening factors $F_{\text{tot,cent}}$, $F_{Y_{2,\text{cent}}}$, and the center value $Z[M]_{Y_{2,\text{cent}}}$ of the transition curve between $Y_{2,0,0}$ and $Y_{2,\infty}$. It appears plausible to assume that the averaging over a thermal J distribution in type-A reactions does not markedly change $Z[M]_{Y_{2,\text{cent}}}$ in comparison to the value for $J = 0$ from eq 16.

On the basis of the foregoing, the parameters $k_{\text{tot},0}$, $k_{\text{tot},\infty}$, and $F_{\text{tot,cent}}$ for k_{tot} as well as $Y_{2,0,0}$, $Y_{2,\infty}$, $Z[M]_{Y_{2,\text{cent}}}$, and $F_{Y_{2,\text{cent}}}$ for Y_2 in type-A reactions are expressed by the corresponding properties of the separated-channel rate constants $k_1^{\text{s.ch.}}$ and $k_2^{\text{s.ch.}}$ (together with the ME result for $k_{2,0}$ from eq 22). Weak collision effects then govern the values of $k_{1,0}^{\text{s.ch.}}$, $k_{2,0}^{\text{s.ch.}}$, and $k_{2,0}$, while angular momentum effects in this class of reactions are suggested to not play a major role.

3.2. Weak Collision and Angular Momentum Effects in Type-B, Two-Channel, Reactions. In type-B, two-channel, reactions, the J dependence of the threshold energies $E_{0i}(J)$ is of central importance. It may, e.g., be approximated by¹⁵

$$E_{01}(J) = E_{01}(J = 0) + (B^\# - B)hcJ(J + 1) \quad (27)$$

for the lower (at $J < J_{\text{sw}}$), rigid-AC, channel 1 and by

$$E_{02}(J) = E_{02}(J = 0) + C_\nu hc[J(J + 1)]^\nu - BhcJ(J + 1) \quad (28)$$

for the upper (at $J < J_{\text{sw}}$), loose-AC, channel 2. $B^\#$, C_ν , and ν depend on the PES of the reaction. For the formaldehyde system, $B = 1.21 \text{ cm}^{-1}$, $B^\# = 1.11 \text{ cm}^{-1}$, $C_\nu \approx 0.43 \text{ cm}^{-1}$, and $\nu \approx 1$ were derived from ab initio calculations of the PES.¹⁵ With $\Delta E_{0i} = E_{02}(J = 0) - E_{01}(J = 0) = 20.1 \text{ kJ mol}^{-1}$, rotational channel switching takes place at $J_{\text{sw}} = 50$ (with $\Delta E_{0i} = 32 \text{ kJ mol}^{-1}$ as used in ref 13, $J_{\text{sw}} = 62$ would be obtained). At $J > J_{\text{sw}}$, channel 2 switches from being the upper channel to being the lower channel. In the simplest way, one then may assume that the rotationally averaged, limiting low-pressure, branching fraction $\langle Y_{2,0,0} \rangle$ is given by that part of the thermal J distribution which belongs to $J > J_{\text{sw}}$. In the prolate symmetric-top approximation, this would be

$$\langle Y_{2,0,0} \rangle \approx [A/(A - B)]^{1/2} \exp[-BhcJ_{\text{sw}}(J_{\text{sw}} + 1)/k_B T] \quad (29)$$

(A and B are the rotational constants of the reactant).

Equation 29 provides a quick estimate for the consequences of rotational channel switching in the J -averaged branching fraction $\langle Y_{2,0,0} \rangle$, but it neglects collisional effects. At $J < J_{\text{sw}}$, the latter would lead to an additional positive contribution to $\langle Y_{2,0,0} \rangle$, arising from collisional up-transitions from energies below to above $E_{02}(J)$. On the other hand, after channel-switching at $J > J_{\text{sw}}$, a contribution due to collisional up-transitions from energies below to above (the now higher) $E_{01}(J)$ has to be subtracted. The latter two contributions do

not cancel but should be of similar magnitude. They can be elaborated by thermal J -averaging the ME result of eq 22. As this requires detailed knowledge of system- and energy-specific branching factors $V_2(E)$ (which also depend on J) as well as rotational effects in the collisions, these refinements do not appear warranted, but for simplicity, one may assume that the mentioned two collisional contributions to $\langle Y_{2,0,0} \rangle$ approximately compensate. Equation 29 then provides a simple estimate for $\langle Y_{2,0,0} \rangle$ in type-B reactions. We validate this assumption for the formaldehyde system in the following section.

After estimating $\langle Y_{2,0,0} \rangle$ by eq 29 and identifying $\langle Y_{2,\infty} \rangle$ with $k_{2,\infty}^{\text{ch.}}/k_{\text{tot},\infty}^{\text{ch.}}$, the center $[M]_{Y_{2,\text{cent}}}$ of the transition of $\langle Y_2 \rangle$ between $\langle Y_{2,0,0} \rangle$ and $\langle Y_{2,\infty} \rangle$ remains to be estimated. Because of rotational channel switching, for type-B reactions, this cannot be done with eq 16. As long as complete ME calculations with J -averaging are not made, a different assumption may be made. We start with separated-channel falloff curves for $k_1^{\text{ch.}}$ and $k_2^{\text{ch.}}$ which are constructed in the standard way.⁷ With appropriate broadening factors, they interpolate between $k_{1,0}^{\text{ch.}}$ and $k_{1,\infty}^{\text{ch.}}$ as well as between $k_{2,0}^{\text{ch.}}$ and $k_{2,\infty}^{\text{ch.}}$. We then assume that $[M]_{Y_{2,\text{cent}}}$ is close to that $[M]$ for which $k_1^{\text{ch.}} \approx k_2^{\text{ch.}}$. We note that rotational channel switching is accounted for in $k_1^{\text{ch.}}$ and $k_2^{\text{ch.}}$, while channel coupling effects in the separated-channel rate constants are neglected. As the importance of the latter effects decreases in the approach to the high-pressure limit, the described procedure to estimate $[M]_{Y_{2,\text{cent}}}$ would appear sufficient. The formaldehyde example (see Figure 3 of ref 13) confirms that the corresponding $[M]_{Y_{2,\text{cent}}}^{\text{ch.}}$ indeed is close to the required $[M]_{Y_{2,\text{cent}}}$. It has to be emphasized that this approach so far is based on plausibility only. Refinements should be derived from more systematic ME calculations for other type-B reaction systems. This is, however, beyond the scope of the present Article.

3.3. Simplified Analysis of the Thermal, Two-Channel, Dissociation of Formaldehyde. The dissociation of formaldehyde, with its rigid-AC channel 1 leading to $\text{CO} + \text{H}_2$ and its loose-AC channel 2 leading to $\text{HCO} + \text{H}$, is a prototype for a type-B, two-channel, reaction system. It has been studied experimentally in detail (see refs 13 and 24 and earlier work cited therein). At the same time, the corresponding photodissociation via the electronic ground state, which is directly related to the thermal dissociation, has also been evaluated.²² In the following, we compare the ME calculations for this reaction from ref 13 with the simplified analysis described in sections 2 and 3.2. Our analysis proposes how the parameters of the representation should be modified when improved experimental and theoretical information becomes available.

We first inspect the performance of eq 10 with respect to the possible need for broadening factors $F(x)$. Figure 1 compares the ME calculations of branching fractions $\langle Y_2 \rangle$ from ref 13 with a representation in the form of eq 10 and a broadening factor $F(x)$ of unity. The derived fit parameters ($\langle Y_{2,0,0} \rangle$, $\langle Y_{2,\infty} \rangle$, $[M]_{Y_{2,\text{cent}}}/10^{-5} \text{ mol cm}^{-3}$) are (0.013, 0.95, 12), (0.08, 0.97, 5.4), and (0.22, 0.98, 1.9) for $T/\text{K} = (1400, 2000, 3000)$, respectively (the given $[M]_{Y_{2,\text{cent}}}$ correspond to $P_{Y_{2,\text{cent}}}/\text{bar} = (12, 8.9, 4.8)$). One observes that the ME results (as shown by the points in Figure 1) can well be fitted in this way. The performance of eq 10 looks similarly good as that by Chebyshev polynomials used in ref 13. However, there are alternatives: values of broadening factors $F(x)$ smaller than

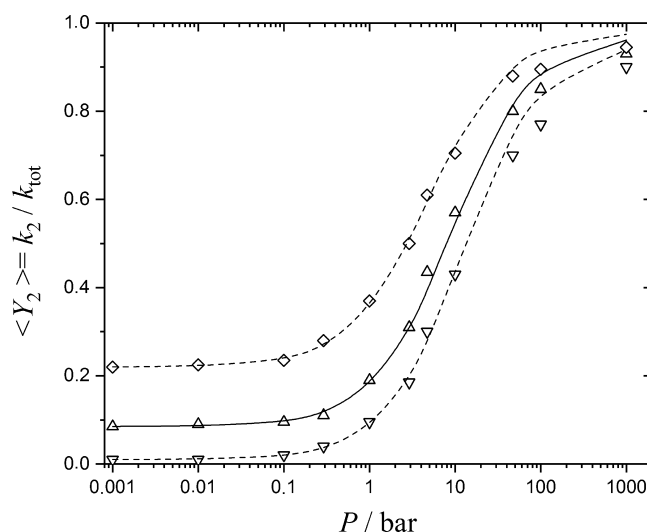


Figure 1. Branching fractions $\langle Y_2 \rangle$ of the loose-activated complex channel $\text{H}_2\text{CO} (+\text{M}) \rightarrow \text{H} + \text{HCO} (+\text{M})$ in the thermal dissociation of formaldehyde (points = master equation results from ref 13 for 1400 (∇), 2000 ($\triangle$), and 3000 K ($\diamond$); lines = representation by eq 10 without broadening factors, from this work; see text).

unity, such as they are typical for falloff curves of $k_1^{\text{ch.}}$, could also be accommodated but would lead to slightly smaller fitted values of $P_{Y_{2,\text{cent}}}$ (broadening factors of $k_1^{\text{ch.}}$, with $F_{1,\text{cent}}^{\text{ch.}} \approx 0.6$ at 2000 K for internal consistency taken from ref 13, e.g., would reduce $P_{Y_{2,\text{cent}}}$ from 8.9 to about 3 bar, without much deteriorating the agreement between the ME points and the fit in Figure 1; the introduction of broadening factors smaller than unity, however, becomes obligatory for larger molecules, such as demonstrated below for the dissociation of glyoxal).

We next analyze the temperature dependence of $\langle Y_{2,0,0} \rangle$ from Figure 1. Over the range 1400–3000 K, the fitted values approximately correspond to

$$\langle Y_{2,0,0} \rangle \approx 2.8 \exp(-7110\text{K}/T) \quad (30)$$

This contrasts with the results from ref 15 which correspond to $\langle Y_{2,0,0} \rangle \approx 0.333 \exp(-871\text{K}/T)$. The difference, however, is understandable, as the two approaches were based on different input parameters. A change of J_{sw} from 62 (as used in ref 13) to 50 (as used in ref 15; see above) according to eq 29 would have changed $\langle Y_{2,0,0} \rangle$ at 2000 K from 0.08 to 0.27, while eq 30 would have led to 0.22. Differences in the average energies transferred per collision (chosen as $\langle \Delta E \rangle/hc = -235 \text{ cm}^{-1}$ in ref 13 and $\langle \Delta E \rangle/hc = -100 \text{ cm}^{-1}$ as derived from thermal dissociation and photodissociation experiments in ref 15) apparently are only of minor importance in type-B reactions. The temperature dependence of $\langle Y_{2,0,0} \rangle$ from the ME calculations of ref 13 is nearly reproduced by eq 29. With $J_{\text{sw}} = 62$, eq 29 leads to $\langle Y_{2,0,0}(T) \rangle / \langle Y_{2,0,0}(2000 \text{ K}) \rangle = (0.22, 1, 3.3)$ for $T/\text{K} = (1400, 2000, 3000)$, while eq 30 leads to (0.16, 1, 3.3). The consequences of changing ΔE_{0i} and J_{sw} thus can well be specified by means of eq 29.

We finally inspect the pressure dependence of the pseudo-second-order rate constants $k_1/[M]$ and $k_2/[M]$. Fits of $Y_1 = 1 - Y_2$ and Y_2 are combined with a simple representation of k_{tot} . Using $k_{\text{tot}} \approx k_{\text{tot},\infty} x/(1+x)$ with $x = P/P_{\text{tot,cent}}$ (and $P_{\text{tot,cent}} \approx 2.6 \times 10^3 \text{ bar}$ from $k_{\text{tot},\infty} = 3.5 \times 10^6 \text{ s}^{-1}$ and $k_{\text{tot},0}/[M] = 2.2 \times 10^8 \text{ cm}^3 \text{ mol}^{-1} \text{ s}^{-1}$ at 2000 K, for consistency taken from ref 13), Figure 2 shows the results. The trends of the ME results

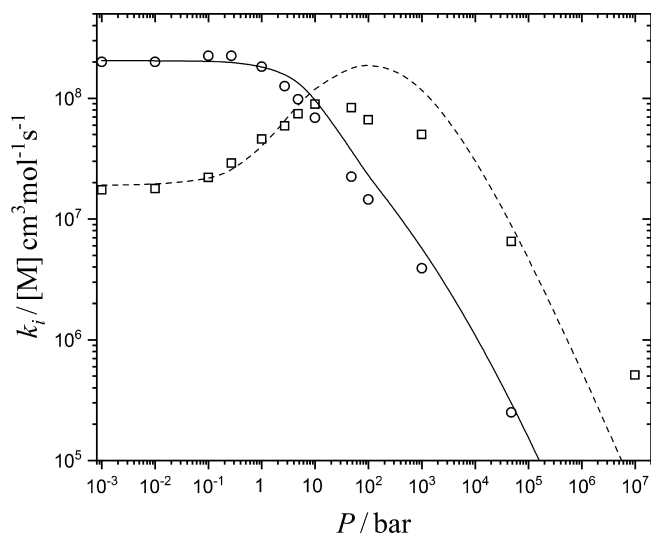


Figure 2. Pseudo-second-order rate constants in the thermal dissociation of formaldehyde at 2000 K (k_1 , rigid-activated complex channel $\text{H}_2\text{CO} (+\text{M}) \rightarrow \text{H}_2 + \text{CO} (+\text{M})$; k_2 , loose-activated complex channel $\text{H}_2\text{CO} (+\text{M}) \rightarrow \text{H} + \text{HCO} (+\text{M})$; $\circ$, master equation results for k_1 from ref 13; $\square$, master equation results for k_2 from ref 13; lines, representation with eq 10 without broadening factors from this work; see text).

are well reproduced, although some deviations are obtained near the maximum of $k_2/[M]$. The latter may be attributed to the omission of broadening factors in $\langle Y_2 \rangle$ and k_{tot} .

3.4. Generalization to Multichannel Dissociation Reactions. While two-channel reactions can be classified in terms of type-A and type-B reactions, the situation becomes more complicated for multichannel reactions. Here each reaction system requires an individual treatment. In spite of this complication, the simplicity of the present approach appears helpful to rationalize the data. We use the dissociation of glyoxal as an illustrative example. It can proceed on three rigid-AC channels with lower threshold energies and one loose-AC channel with higher threshold energy (see the Introduction). Detailed experiments and ME calculations for this system have been described in ref 19. The results in the following provide the basis for a simplified analysis and data representation, analogous to that described for formaldehyde in section 3.3.

Figure 3 shows ME results for the branching fractions $\langle Y_i \rangle$ at 1300 K of the four channels ($i = 1a$ for the products $\text{CH}_2\text{O} + \text{O}$, $1b$ for the products $2\text{CO} + \text{H}_2$, and $1c$ for the products $\text{HCOH} + \text{CO}$ and $i = 2$ for the products 2HCO). A comparison of the total branching fraction of the rigid-AC channels $\langle Y_{1\text{tot}} \rangle = \langle Y_{1a} \rangle + \langle Y_{1b} \rangle + \langle Y_{1c} \rangle$ with that of the loose-AC channel $\langle Y_2 \rangle$ shows a crossover at $[M] \approx 5 \times 10^{-4} \text{ mol cm}^{-3}$. This is a clear indication for rotational channel switching between the three rigid-AC channels and the loose-AC channel, similar to that described for the formaldehyde system. A representation of $\langle Y_2 \rangle$ in the form of eq 10 at the given temperature could be attempted with $\langle Y_{2,0,0} \rangle \approx 0$ and a value of $\langle Y_{2,\infty} \rangle$ larger than 0.5. A closer inspection of Figure 3, however, shows that eq 10 without introducing broadening factors would correspond to a much too rapid transition from $\langle Y_{2,0,0} \rangle$ to $\langle Y_{2,\infty} \rangle$. RRKM calculations for the dominant rigid-AC channel 1a (done with the molecular parameters given in the Supporting Information of ref 19) lead to $F_{1a,\text{cent}}^{\text{ch}} = 0.34$.

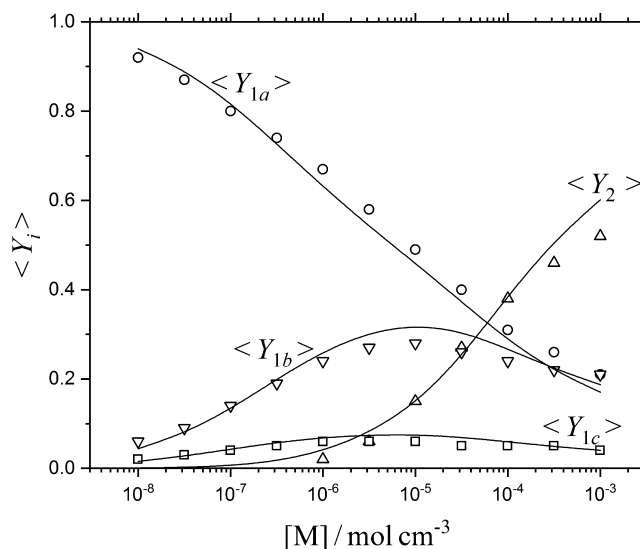


Figure 3. Branching fractions $\langle Y_2 \rangle$ for the loose-activated complex channel $(\text{CHO})_2 (+\text{M}) \rightarrow 2\text{HCO} (+\text{M})$ and the rigid-activated complex channels $(\text{CHO})_2 (+\text{M}) \rightarrow \text{CH}_2\text{O} + \text{CO} (+\text{M})$ ($\langle Y_{1a} \rangle$), $(\text{CHO})_2 (+\text{M}) \rightarrow 2\text{CO} + \text{H}_2 (+\text{M})$ ($\langle Y_{1b} \rangle$), and $(\text{CHO})_2 (+\text{M}) \rightarrow \text{HCOH} + \text{CO} (+\text{M})$ ($\langle Y_{1c} \rangle$) in the thermal dissociation of glyoxal (points = master equation results from ref 19, lines = empirical fits with eqs 31–34 from this work; see text).

We tentatively include the corresponding broadening factor in a fit of $\langle Y_2 \rangle$ by eq 10. $\langle Y_2 \rangle$ then is empirically represented by

$$\langle Y_2 \rangle \approx \langle Y_{2,\infty} \rangle [x_2 / (1 + x_2)] F_2(x_2) \quad (31)$$

with $x_2 = [M]/[M]_{2,\text{cent}}$ and $\log F_2(x_2) \approx [1 + ((\log x_2)/N)^2]^{-1} \log F_{2,\text{cent}}$, where $N \approx 0.75 - 1.27 \log F_{2,\text{cent}}$; see above. RRKM calculations¹⁹ of the limiting high-pressure rate constants at 1300 K led to $\langle Y_{2,\infty} \rangle \approx 0.85$. This places $[M]_{2,\text{cent}}$ at that $[M]$ where the ME result for $\langle Y_2 \rangle$ is equal to $0.85 \times 0.5 \times 0.34 \approx 0.15$, i.e., at $[M]_{2,\text{cent}} \approx 10^{-5} \text{ mol cm}^{-3}$. Figure 3 shows that ME results and their empirical fit by eq 10 with the broadening factor from k_{1a} reasonably agree.

After having represented the type-B channel $i = 2$ by its branching fraction $\langle Y_2 \rangle$, the remaining branching fraction $1 - \langle Y_2 \rangle$ has to be partitioned among the three type-A channels $i = 1a, 1b$, and $1c$. One could do this with the procedure described in section 3.2, treating channels $1b$ and $1c$ individually as upper channels relative to the lowest channel $1a$. At this stage, however, an empirical representation of the ME results appears more warranted, in particular as empirical broadening factors are introduced into the representation. With empirically fitted $\langle Y_{1b,\infty} \rangle \approx 0.54$ and $\langle Y_{1c,\infty} \rangle \approx 0.11$ at 1300 K,

$$\langle Y_{1b} \rangle \approx (1 - \langle Y_2 \rangle) \langle Y_{1b,\infty} \rangle [x_{1b} / (1 + x_{1b})] F_{1b}(x_{1b}) \quad (32)$$

$$\langle Y_{1c} \rangle \approx (1 - \langle Y_2 \rangle) \langle Y_{1c,\infty} \rangle [x_{1c} / (1 + x_{1c})] F_{1c}(x_{1c}) \quad (33)$$

$$\langle Y_{1a} \rangle \approx (1 - \langle Y_2 \rangle - \langle Y_{1b} \rangle - \langle Y_{1c} \rangle) \quad (34)$$

are calculated. Like $\langle Y_{2,0,0} \rangle$, also the low-pressure limiting $\langle Y_{1b,0,0} \rangle$ and $\langle Y_{1c,0,0} \rangle$ are neglected (see below). $F_{1b,\text{cent}} \approx 0.23$ and $F_{1c,\text{cent}} \approx 0.28$ are taken from RRKM calculations, such that the centers of the transition curves can be located. One obtains $[M]_{1b,\text{cent}} \approx 2 \times 10^{-8} \text{ mol cm}^{-3}$ and $[M]_{1c,\text{cent}} \approx 10^{-8} \text{ mol cm}^{-3}$. Besides the results for $\langle Y_2 \rangle$, Figure 3 also shows reasonable agreement between the ME results for the channels $1a, 1b$, and $1c$ and their empirical representation by eqs 31–34.

The absolute value of $\langle Y_{2,0,0} \rangle$ for the type-B channel $i = 2$ and its temperature dependence can be analyzed with eq 29. Using $E_{01a}(J) \approx E_{01a}(J = 0) = 290.9 \text{ kJ mol}^{-1}$, $E_{02}(J = 0) = 231.8 \text{ kJ mol}^{-1}$, and $B = 0.152 \text{ cm}^{-1}$ from ref 19 and putting $C_v \approx 0$, eqs 27 and 28 lead to $J_{sw} \approx 180$ for rotational channel switching. With eq 29, this gives $\langle Y_{2,0,0} \rangle \approx 4.2 \times 10^{-3}$ at 1300 K and 5.8×10^{-2} at 2500 K. The neglect of $\langle Y_{2,0,0} \rangle$ in eq 31 for 1300 K thus appears justified. For type-A channels, instead of eq 29, eq 24 is used for the estimate of $\langle Y_{1b,0,0} \rangle$ and $\langle Y_{1c,0,0} \rangle$. Because the threshold energies of the type-A channels 1b and 1c are about 2000 cm^{-1} larger than the threshold energy of channel 1a and γ is certainly much smaller than 2000 cm^{-1} , the neglect of $\langle Y_{1b,0,0} \rangle$ and $\langle Y_{1c,0,0} \rangle$ under all conditions also appears justified. The temperature dependences of $F_{1b,cent}$ and $F_{1c,cent}$ as well as of $\langle Y_{1b,\infty} \rangle$ and $\langle Y_{1c,\infty} \rangle$ relative to the values at 1300 K are tentatively taken from RRKM calculations (in spite of the fact that the fitted $\langle Y_{1b,\infty} \rangle$ and $\langle Y_{1c,\infty} \rangle$ do not correspond to the RRKM results of ref 19). Without going into further details, one may assume in addition that the temperature dependences of $[M]_{1b,cent}$ and $[M]_{1c,cent}$ correspond to those of the related separated-channel quantities. We note that the temperature dependences of the channel branching fractions at fixed bath gas concentrations $[M]$, such as those illustrated in ref 19, are reproduced reasonably well.

The dissociation of glyoxal with three rigid-AC channels and one loose-AC channel represents only one possible example for multichannel behavior. In a separate publication, multichannel coupling effects in the thermal dissociation of CH_3F will be analyzed (see the Introduction). The situation here is different: the lowest, spin-forbidden, dissociation leading to $^3\text{CH}_2 + \text{HF}$ assumes the role of the rigid-AC channel, while the higher dissociation to $\text{CH}_2\text{F} + \text{H}$ is a loose-AC bond fission channel. On the other hand, the intermediate channel leading to $^1\text{CH}_2 + \text{HF}$ through a weakly bound $^1\text{CH}_2\text{--HF}$ intermediate is of intermediate rigid/loose-AC channel character which requires a treatment different from those described above. This example illustrates the necessity to provide an individual treatment for each multichannel system, although one may follow the lines sketched for the glyoxal system.

4. CONCLUSIONS

The complicated intrinsic dynamics of multichannel thermal unimolecular dissociations in general calls for master equation treatments in the framework of unimolecular rate theory. The influence of specific molecular input parameters then could be explored by their systematic variation in the ME calculations. As this may become an involved task, however, a simplified modeling approach such as that presented in this Article also appears justified. It emphasizes the marked differences in the channel coupling between rigid-/rigid- and rigid-/loose-activated complex channel systems. It suggests that the former are governed by weak-collision, energy-transfer, effects, while the latter are dominated by angular-momentum, rotational channel switching, effects. The present approach suggests how separated-channel rate constants neglecting channel coupling can be converted to coupled-channel rate constants accounting for the coupling. It finally provides approximate rate constant representations which appear suitable for inclusion in databases. As only the dissociations of formaldehyde and of glyoxal have been used as test examples, clearly more work is needed to validate the drawn conclusions.

AUTHOR INFORMATION

Corresponding Author

*E-mail: shoff@gwdg.de.

ORCID

Jürgen Troe: 0000-0002-4347-0307

Notes

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